

Where service gaps overlap

A national screen of quality, accessibility, population-adjusted coverage and operational capacity.

Perspective: a data scientist supporting ACECQA and government decision makers



17,663 approved services	99.9% matched to ABS remoteness	4 transparent screening rules	0-13 population denominator
-----------------------------	------------------------------------	----------------------------------	--------------------------------

KEY BUSINESS QUESTION

The decision problem is overlap, not a single ranking

Where do gaps in service quality, accessibility and operational capacity overlap across Australia, and which geographic groups warrant closer investigation?

01

Quality

Below-NQS share and seven quality areas

02

Accessibility

Bus and train proximity, kept separate

03

Coverage

Approved places per 1,000 children

04

Operations

Centre-Based capacity and annual hours

OBSERVE —> COMPARE —> ADJUST —> SCREEN —> VALIDATE LOCALLY

Outcome: a defensible investigation screen, not a causal model or automatic funding allocation.

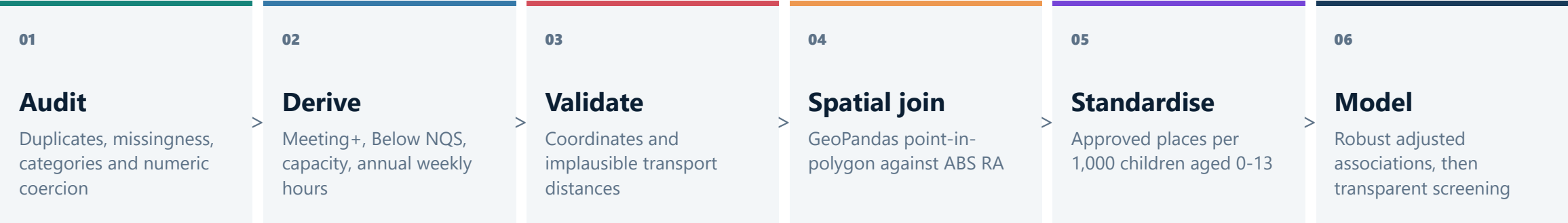
DATA SOURCES

Three sources turn a register into a geographic planning dataset

01	ACECQA approved services register Service type, ratings, approval details, capacity and operating calendars.	17,663 service records			
02	Instructor-enriched location fields Coordinates, nearest bus and train station, and distance to each station.	114 >500 km records reviewed			
03	ABS ASGS 2021 and Census DataPack Official Remoteness Area polygons and exact ages 0-13 usual-resident counts.	4.32m children in scope			
<table><tr><td>UNIT OF ANALYSIS Approved service</td><td>GEOGRAPHIC JOIN State-specific RA code</td><td>REGISTER DESIGN Cross-sectional snapshot</td></tr></table>			UNIT OF ANALYSIS Approved service	GEOGRAPHIC JOIN State-specific RA code	REGISTER DESIGN Cross-sectional snapshot
UNIT OF ANALYSIS Approved service	GEOGRAPHIC JOIN State-specific RA code	REGISTER DESIGN Cross-sectional snapshot			

Sources: [ACECQA National Registers](#); [ABS ASGS Edition 3](#); [ABS 2021 Census DataPacks](#).

Preparation preserves the meaning of ratings, calendars and geography



Analytical safeguards

No invented quality score. Ratings remain ordered categories.

No calendar summation. Annual, school-term and holiday schedules remain separate.

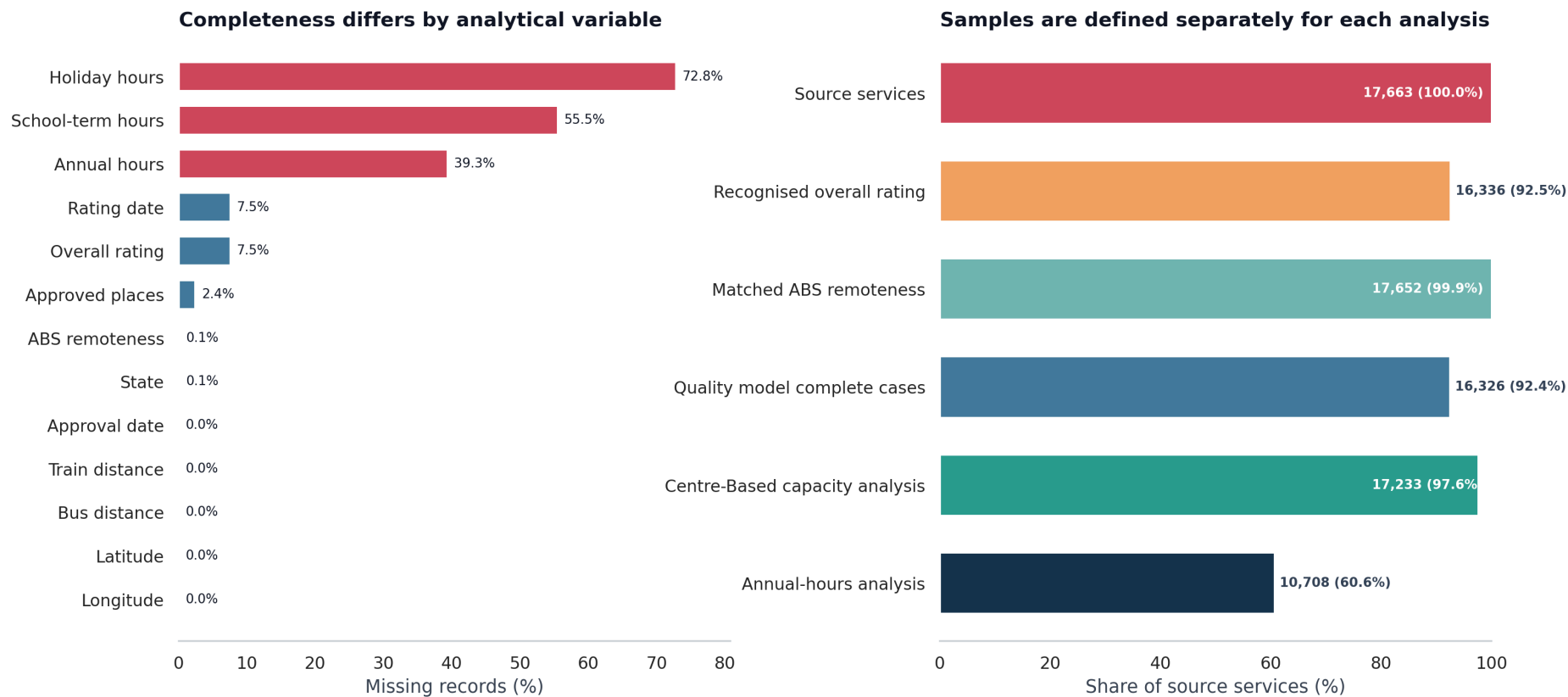
No global row deletion. Each analysis uses its own valid denominator.

No capacity comparison for Family Day Care. Approved places are not comparable.

0 duplicate service IDs	841 overlapping sessions merged	17,652 services matched to RA
-----------------------------------	---	---

Missingness changes the denominator more than the map

EDA 1 | Data readiness and analysis-specific denominators



Rows are not globally deleted for one missing field. Red completeness bars exceed 20% missing; each analysis uses only the variables it requires.

Rows are retained for analyses where their required variables are valid; missing hours do not remove a service from geographic or quality analysis.

92.5%

recognised overall rating

39.3%

annual hours missing

60.6%

annual-hours model sample

Every percentage and model reports its own valid *n*.

Right-skew and service structure determine the visual and model choices

MEASURE	SCOPE	VALID N	MISSING	MEAN	MEDIAN	P90
Bus distance	All services	17,663	0.0%	32.1 km	10.2 km	76.2 km
Train distance	All services	17,663	0.0%	11.4 km	2.5 km	20.6 km
Nearest listed transport	All services	17,663	0.0%	9.5 km	2.3 km	16.0 km
Approved places	Centre-Based	17,244	0.0%	68.8	60.0	120.0
Annual hours / week	All services	10,716	39.3%	54.0 h	55.0 h	60.0 h

DISTRIBUTION

Use medians, P90 and log distance

Means are pulled upward by the long transport tail.

COMPARABILITY

Restrict capacity to Centre-Based Care

Family Day Care lacks a comparable approved-place measure.

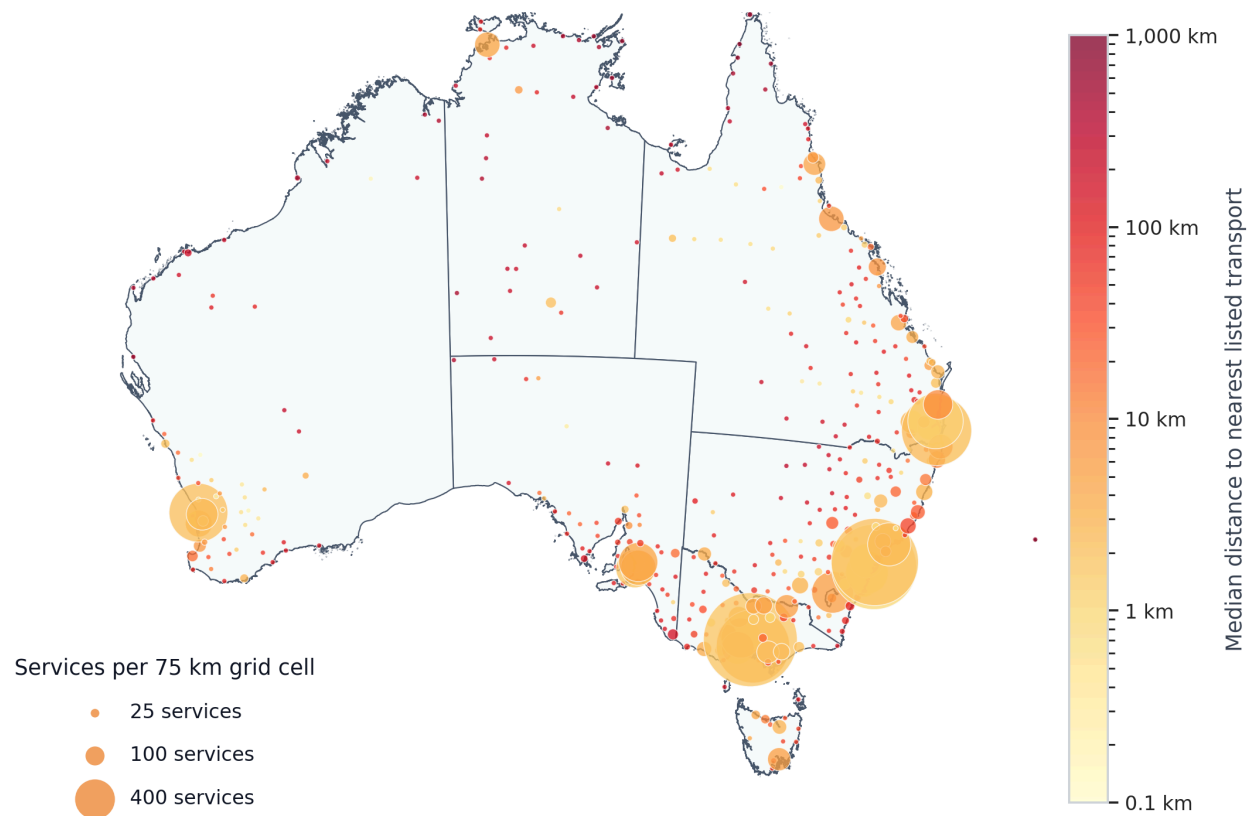
CONFOUNDING

Control state, remoteness and broad type

Sample composition changes substantially across geography.

Approved services cluster along coastal urban corridors

Approved services are concentrated along coastal urban corridors



Observed provision only: bubble area represents registered services, not places per child or unmet demand. ABS 2021 state and remoteness boundaries.

READ THIS AS

Observed provision

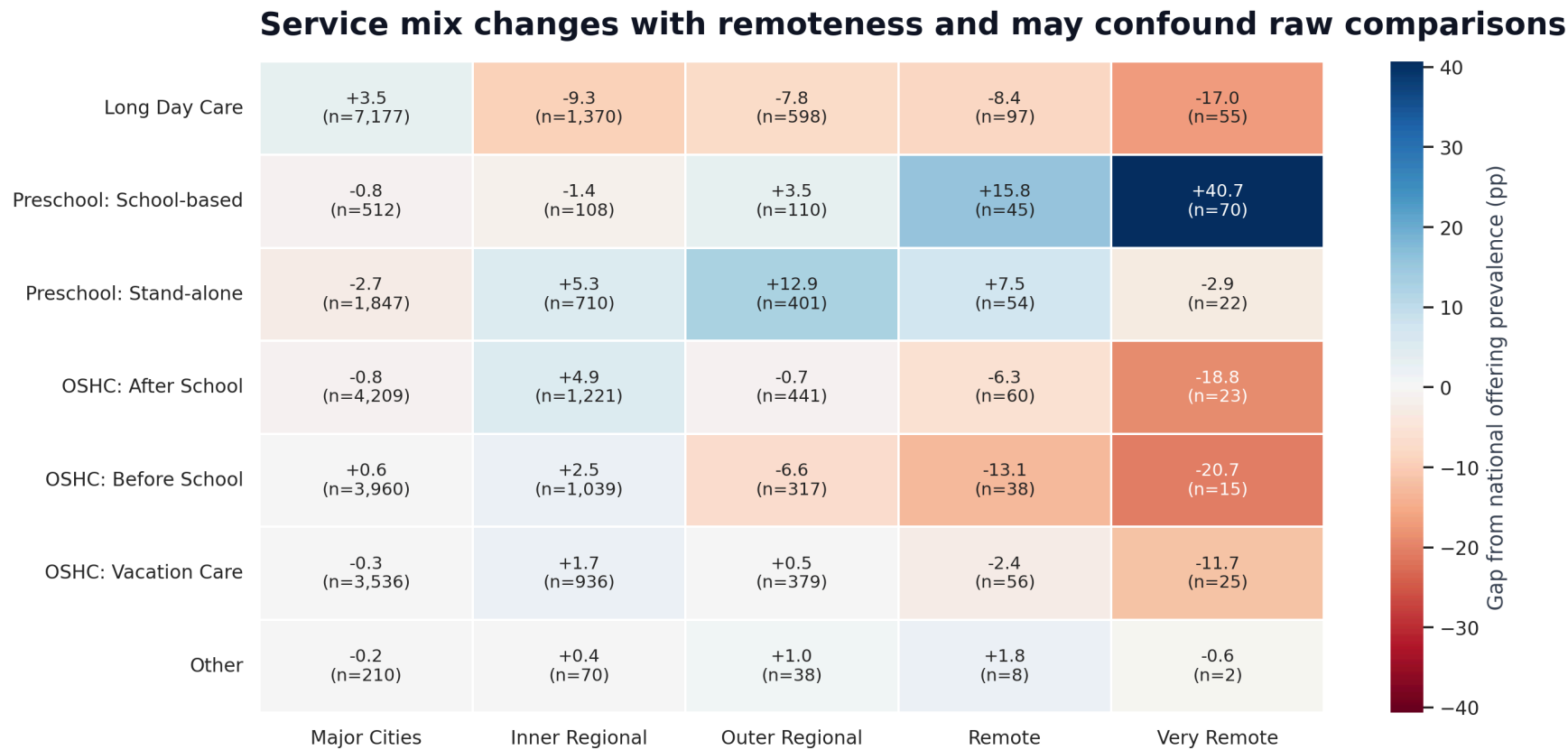
Bubble area is service count in a 75 km equal-area cell.

Colour is median distance to the nearest listed transport mode.

Counts alone do not show unmet need.

[Open live explorer](#)

Service composition changes with remoteness



Cells show percentage-point differences from each offering's national prevalence; counts are included to expose small remote samples.

LARGEST SHIFT

+40.7 pp

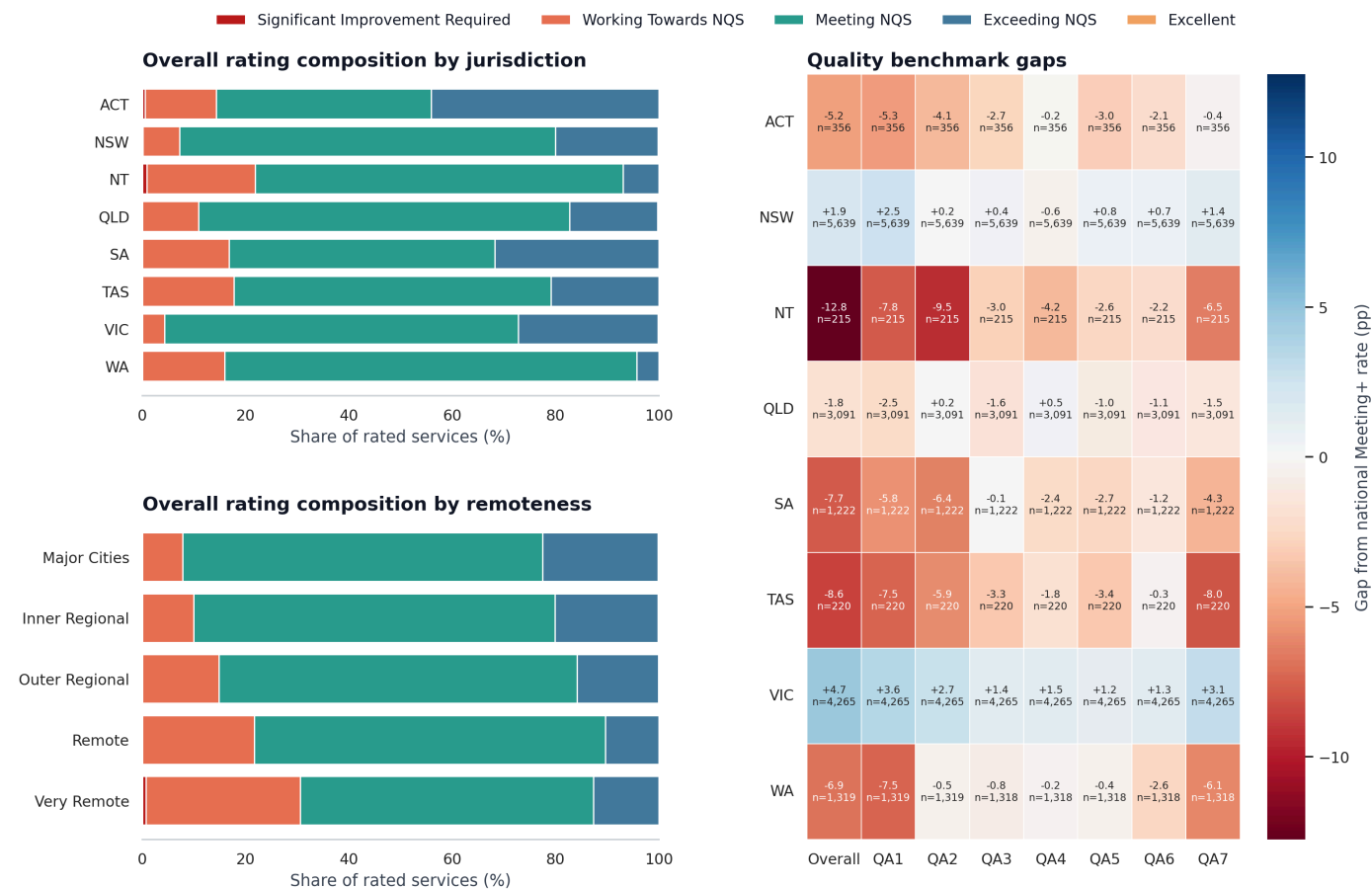
School-based preschool prevalence in Very Remote Australia versus its national prevalence.

Offerings overlap; cells are not market shares.

Implication: raw geographic quality and hours comparisons can partly reflect a different service mix.

Quality differences are geographic, but raw comparisons are compositional

Quality outcomes vary across jurisdictions and remoteness groups



Compositions use recognised ratings only. Benchmark cells show percentage-point gaps from the national Meeting+ rate and the rated denominator n.

21.9%

NT Below NQS among rated services

30.6%

Very Remote Below NQS

QA1

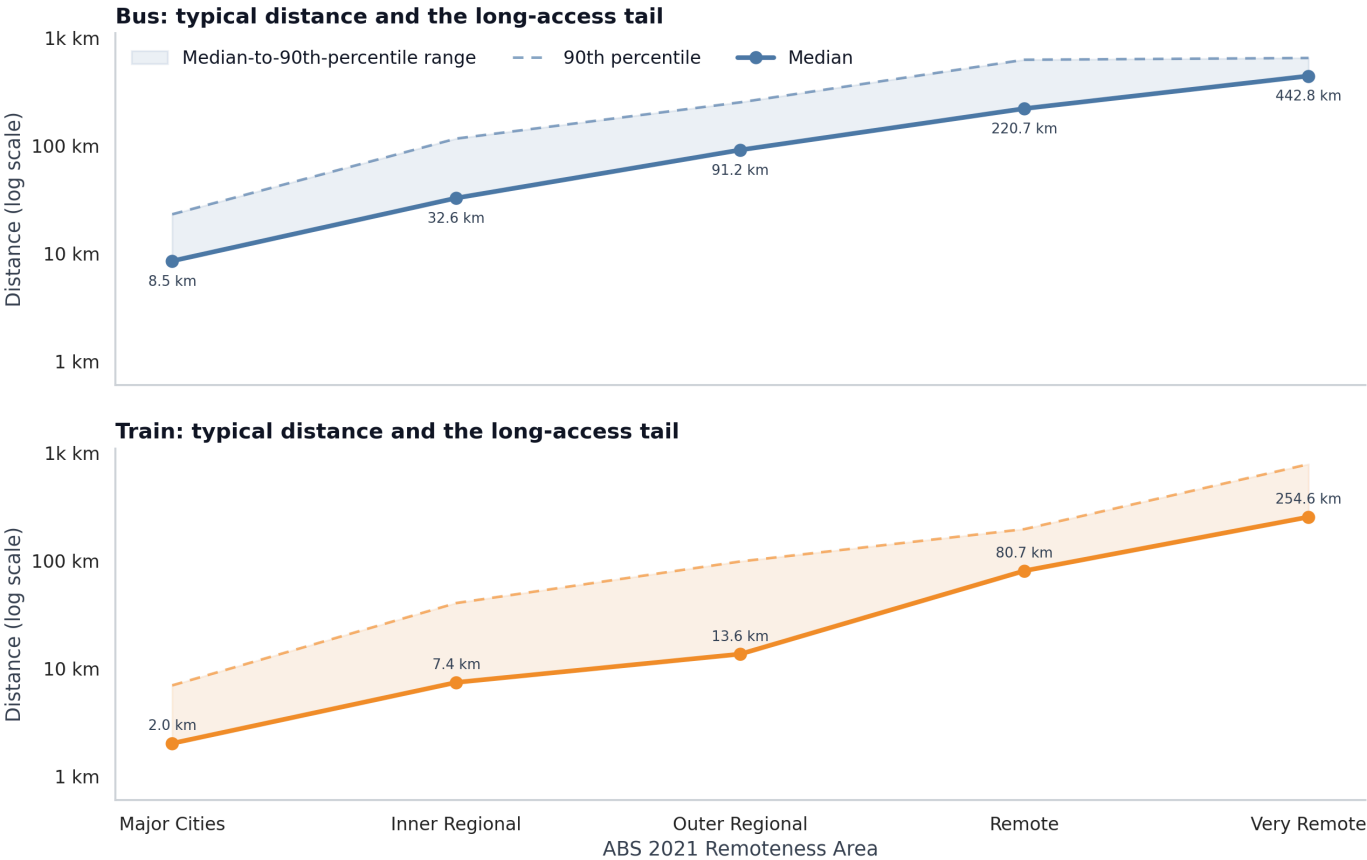
highest national Below-NQS area:
6.2%

Recognised ratings only;
benchmark cells include rated n.

Explore quality

Transport distance rises rapidly with remoteness

Transport distance grows rapidly with remoteness, including its long tail



Lines use all services with valid distance data. The shaded range from the median to P90 exposes skew without treating distance as travel time, frequency, affordability, or actual family behaviour.

NEAREST LISTED MODE

1.89 → 230.03
km Major Cities km Very Remote

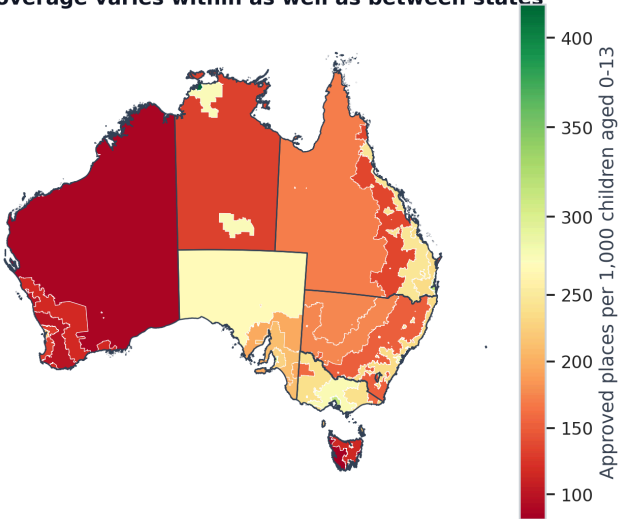
Bus and train remain separate because no rail is not the same as no transport.

Distance is not travel time, frequency or affordability.

Population adjustment reveals lower approved-place intensity outside major cities

Population adjustment reveals lower approved-place intensity outside major cities

Coverage varies within as well as between states



State x remoteness benchmark | national = 275

	Major Cities	Inner Regional	Outer Regional	Remote	Very Remote
ACT	401	140*			
NSW	279	233	151	175*	175*
NT			418	277	132
QLD	319	235	245	135	170
SA	308	228	211	196	271*
TAS		191	114	81*	214*
VIC	316	279	238	158*	
WA	215	130	99	116	87

APPROVED PLACES / 1,000 CHILDREN

293.9

Major Cities

↓

141.2

Very Remote

National benchmark: 274.6

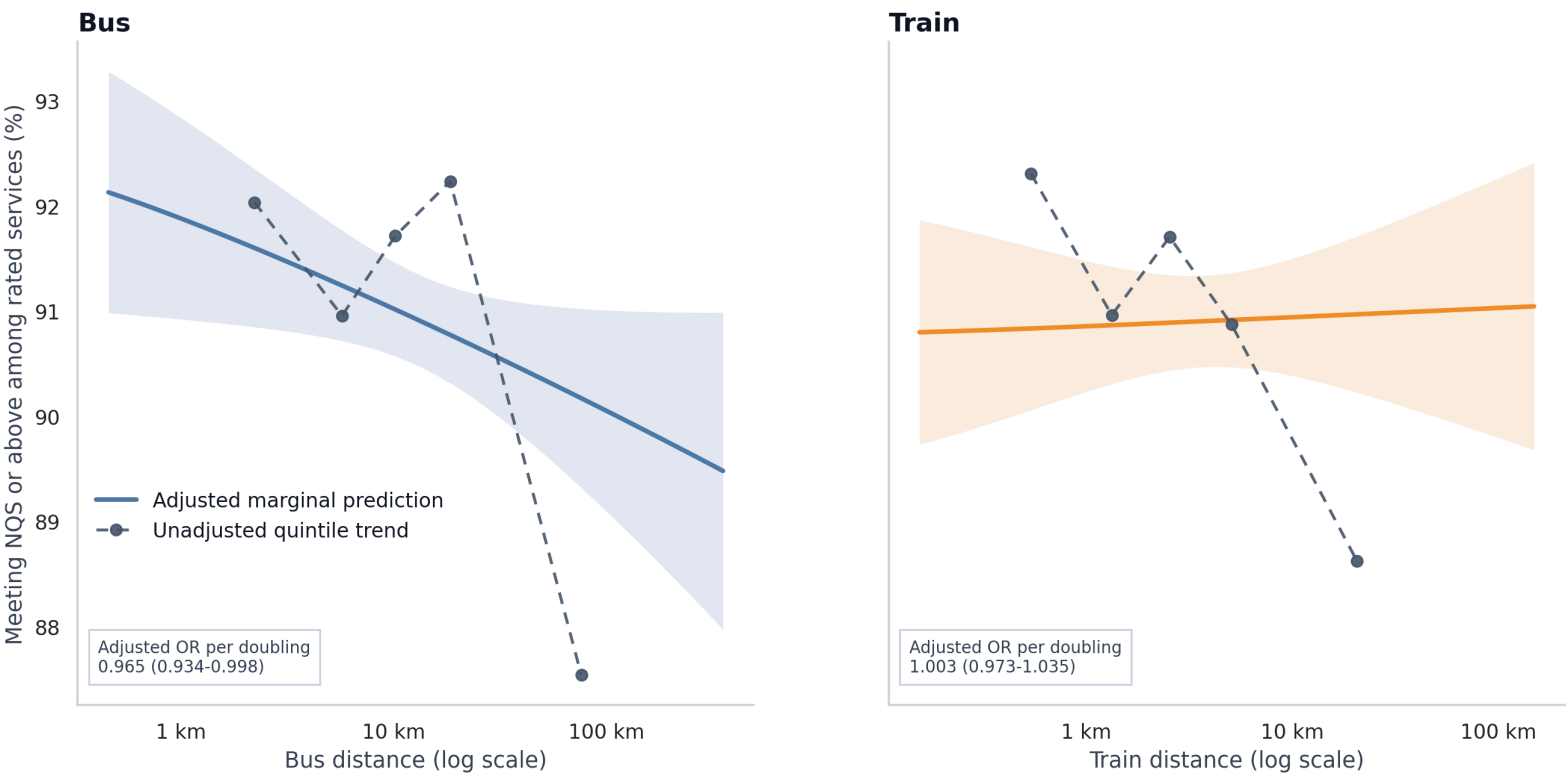
Places are not vacancies, enrolments or demand.

Explore coverage

Numerator: current-register Centre-Based approved places. Denominator: ABS 2021 Census usual residents aged 0-13. Red cells are below the national benchmark; * marks fewer than 5,000 children. Places are not vacancies and the dates differ.

Adjustment substantially weakens the raw quality-accessibility gradient

Adjustment substantially attenuates the raw quality-accessibility gradient



Binomial GLMs control for remoteness, state, and broad service type. Curves average predictions over the observed covariate distribution; raw-quintile uncertainty is retained in Appendix A03. Associations are not causal.

Most of the raw gradient is geographic or compositional; transport distance alone is not a sufficient quality intervention target.

Bus distance doubling

OR 0.965

95% CI 0.934-0.998

Train distance doubling

OR 1.003

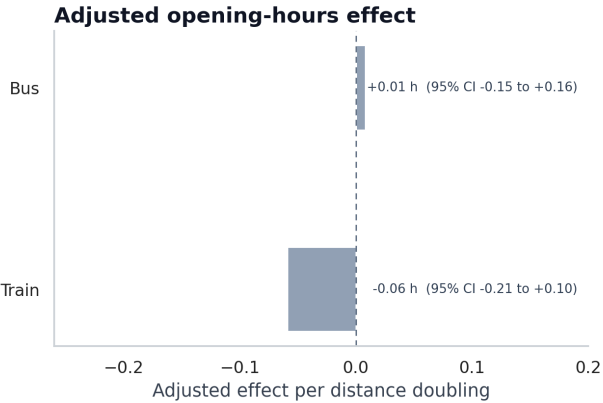
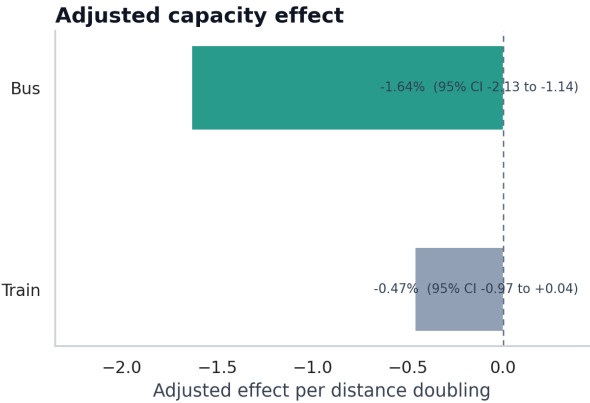
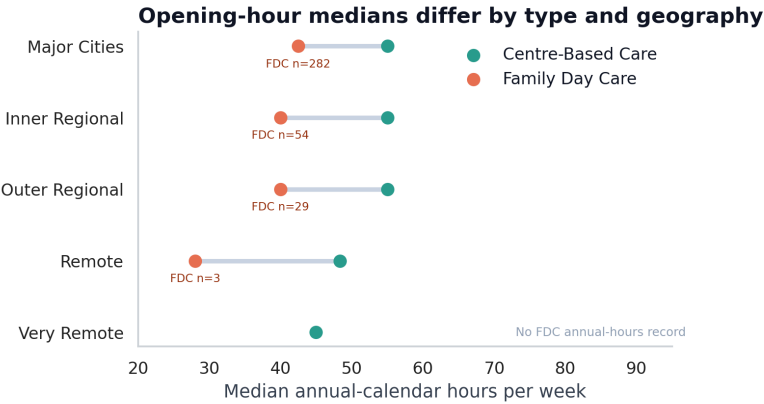
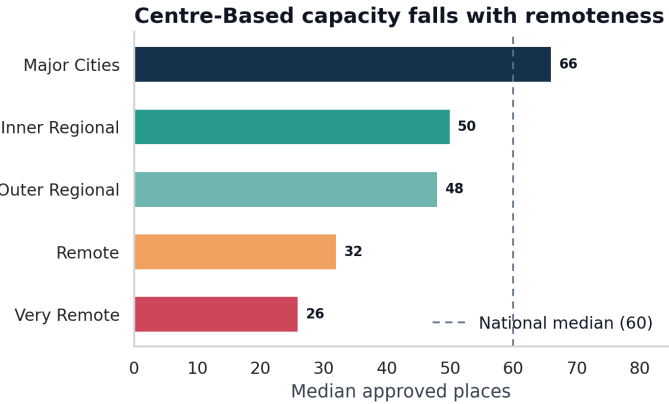
95% CI 0.973-1.035

Controls: remoteness, state and broad service type.

Association is not causation.

Capacity declines with remoteness; opening-hours evidence is weaker

Lower remote capacity is visible; opening-hours evidence is less consistent



Capacity is Centre-Based only. Family Day Care hour medians have very small remote samples. Models control remoteness and state; hours also control broad type. Robust confidence intervals are printed beside each effect.

66
places
Major Cities

→

26
places
Very Remote

Bus distance doubling

-1.64% capacity

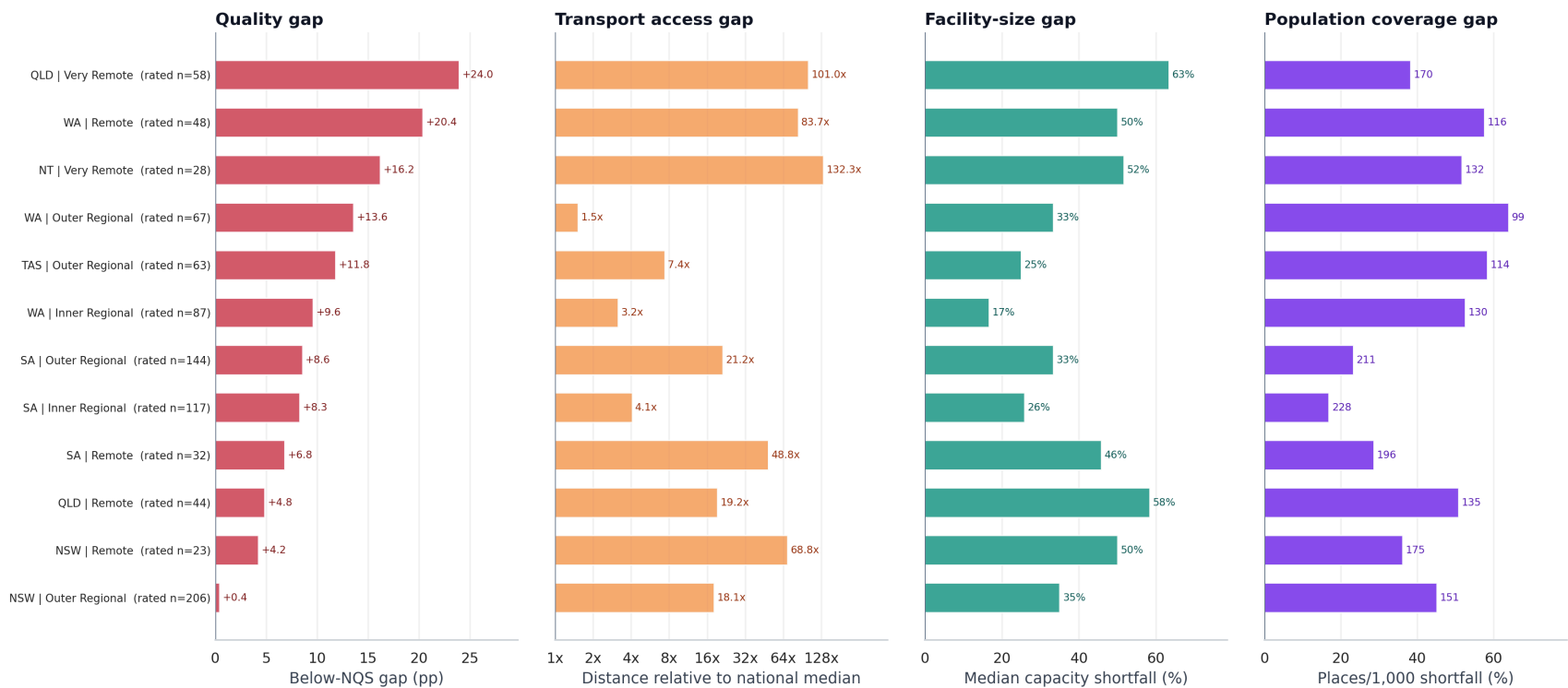
95% CI -2.13 to -1.14

Adjusted annual-hours effects are near zero for both modes.

Hours use annual-calendar records only.

Twelve geographic groups cross all four national benchmarks

12 geographic groups cross all four screening benchmarks



Centre-Based Care only. Every row has above-benchmark Below-NQS and transport distance, below-benchmark median facility size, and below-benchmark approved places per 1,000 children. Coverage labels show places per 1,000; no weighted score is created.

Centre-Based State × Remoteness groups with at least 20 rated services. No weighted score is created.

- 1 Below-NQS rate above national
- 2 Transport distance above national median
- 3 Median capacity below national
- 4 Places per child below national

Inspect all groups

INTERPRETATION BOUNDARIES

The screen is useful only when its limitations stay visible

TIME Snapshot, not trend A current register cannot establish changes or causes.	DENOMINATOR 2021 versus current Census population and the service snapshot are not contemporaneous.	NEED Places are not demand No enrolment, vacancy, waitlist or staffing data are available.
ACCESS Distance is a proxy No travel time, frequency, cost, route quality or car access.	POPULATION Age fit is broad Ages 0-13 do not match every offering's eligible population.	GEOGRAPHY Aggregation hides variation RA groups can conceal local pockets and small-cell uncertainty.

DECISION GUARDRAIL

Use the flags to commission local validation, not to label communities or allocate resources directly.

Turn the screen into three policy decisions

NOW Commission joint local reviews Validate QA1 standards, rated samples, utilisation, waitlists and realistic journeys in the 12 groups. OWNER: ACECQA + STATE REGULATORS	NEXT Match action to the diagnosed gap Use support or regulatory follow-up for quality; test flexible provision or transport coordination where access need is confirmed. OWNER: ACECQA + GOVERNMENTS	EMBED Monitor national consistency Refresh the four indicators and track whether quality and access gaps narrow across jurisdictions. OWNER: ANALYTICS AND REPORTING
--	---	--

IMMEDIATE DECISION

Authorise the 12-group review and require a local evidence brief before intervention.

CONCLUSION AND DECISION

Three decisions for ACECQA and governments

Very Remote Australia combines a 230 km median distance, about 141 approved places per 1,000 children and smaller centres. Twelve groups cross all four screening benchmarks, while adjustment shows that distance alone does not explain quality.

Quality

Target guidance, training or regulatory follow-up at confirmed QA1 weaknesses

Access

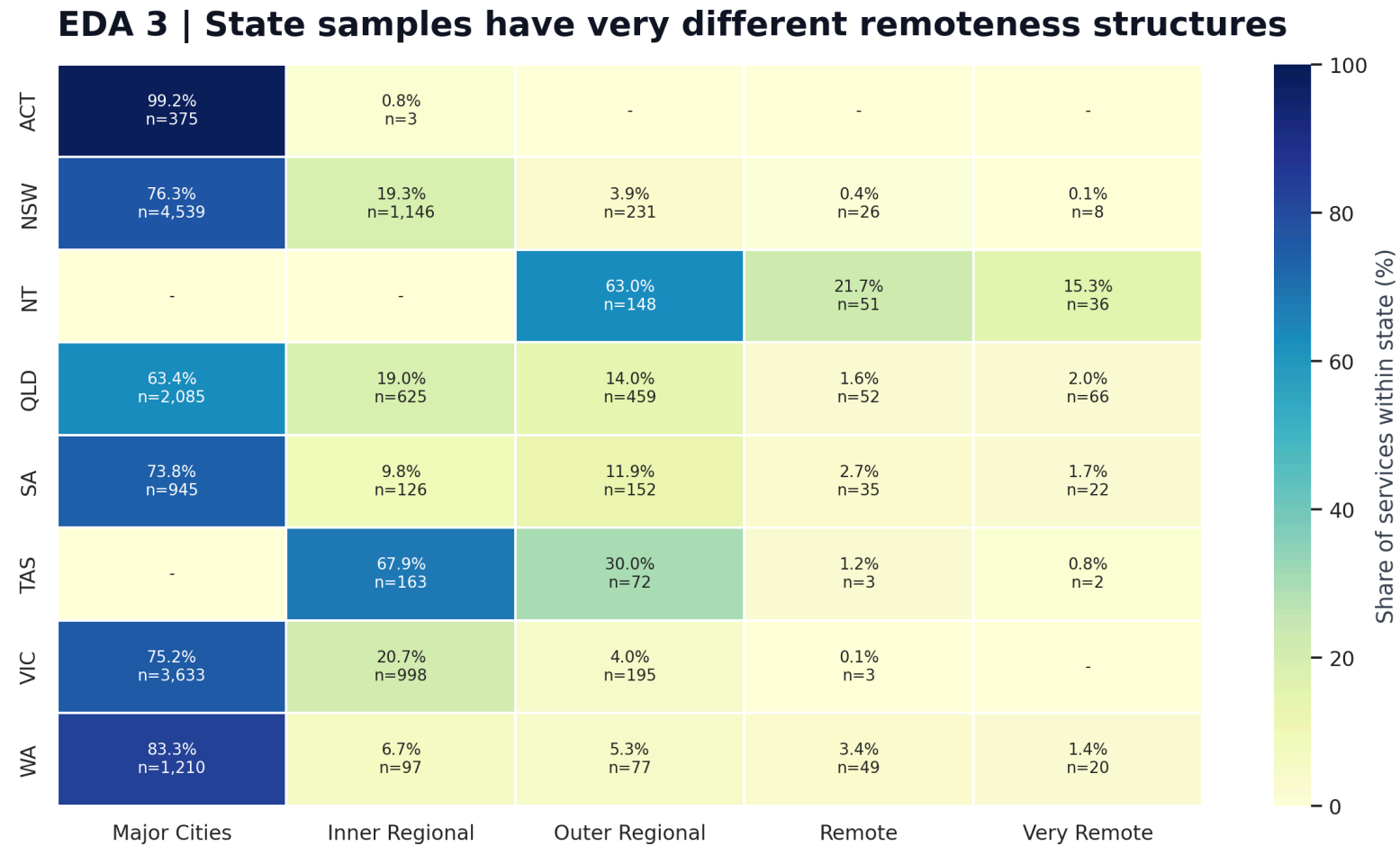
Use current demand and journey evidence to choose local service or transport responses

Governance

Refresh the screen with jurisdictions and monitor consistency under the National Law

Prioritise the 12 groups for evidence and tailored support, not automatic funding or enforcement.

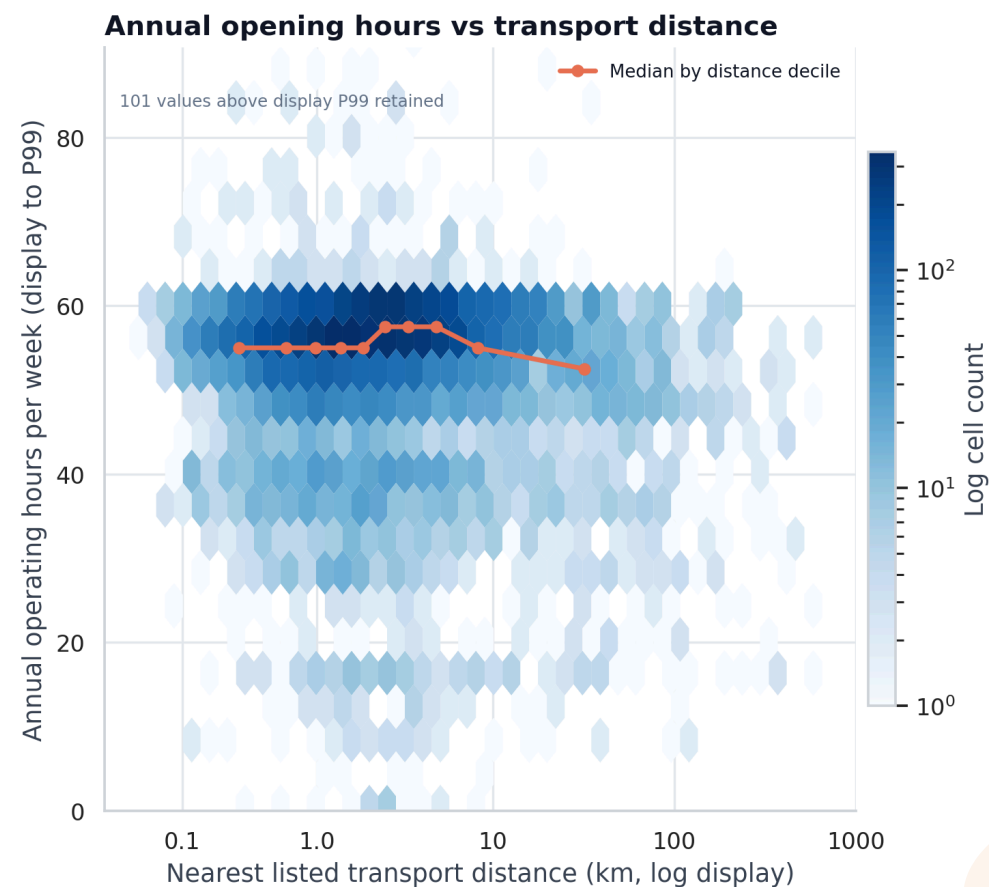
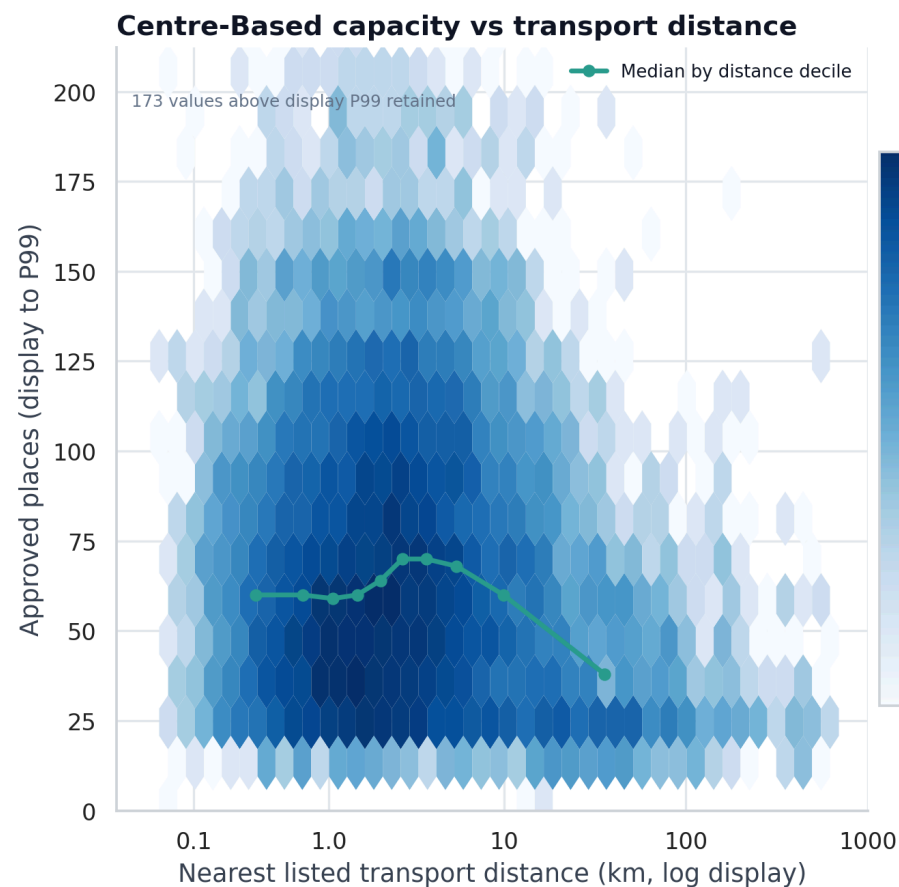
State samples have different remoteness structures



Percentages are calculated within each state; counts identify sparse Remote and Very Remote cells that require uncertainty intervals or minimum-n rules.

Raw operational relationships require adjustment

EDA 4 | Raw operational relationships require geographic adjustment



Hexagons reduce overplotting; trend lines show unadjusted medians only. These patterns motivate models controlling for remoteness, state, and service type and must not be interpreted causally.

Adjusted model specifications

QUALITY

Binomial GLM + HC1

```
Meeting+ ~ log2(distance + 0.1) + remoteness +  
service type + state
```

n = 16,326

CAPACITY

Log-capacity OLS + HC3

```
log(places) ~ log2(distance + 0.1) + remoteness  
+ state
```

Centre-Based n = 17,233

HOURS

Annual-hours OLS + HC3

```
hours ~ log2(distance + 0.1) + remoteness +  
service type + state
```

n = 10,708

INFERENCE

Robust intervals quantify uncertainty around adjusted associations; they do not identify causal effects.

Supporting analyses retained outside the core story

